

POST-PCI HYPOTENSION

Index case for cross-domain evidence retrieval

Patient	75-year-old woman
Procedure	Complex RCA CTO PCI with dual access
Signal	Abrupt refractory hypotension with elevated venous pressure
Scope	Clinical state through the decision to obtain urgent CT

A 75-year-old woman with treated hypertension and long-standing stable angina despite maximal medical therapy (bisoprolol 5 mg daily, amlodipine 5 mg daily, isosorbide mononitrate 40 mg twice daily) was referred for PCI of a chronic total occlusion (CTO) in the proximal right coronary artery (RCA). Baseline ECG showed sinus rhythm without Q waves, and transthoracic echocardiography revealed preserved LV systolic function with mild inferior hypokinesia. Coronary angiography confirmed a dominant RCA CTO (J-CTO score 1) and non-flow-limiting disease in the left system.

The procedure was performed via dual access (right radial 6 Fr and right femoral 8 Fr). After crossing the occlusion with a Gaia 2nd wire and pre-dilating with a 2.5 x 15 mm balloon, extensive dissection occurred. Two overlapping drug-eluting stents (3.5 x 36 mm and 3.5 x 33 mm) were deployed, after which her blood pressure fell to 60/40 mmHg. Two additional overlapping stents were placed distally, but hypotension persisted, refractory to fluid boluses and intravenous phenylephrine boluses. The patient complained of dizziness and had elevated venous pressures.

Pericardial tamponade was suspected; an urgent echocardiogram showed no pericardial effusion in the subcostal view. A careful review of the angiograms showed no evidence of coronary perforation or catheter-related dissection in the left system, and the ECG remained unchanged. Femoral angiography excluded access-site bleeding. Other considerations - nitrate effect, vagal reaction, and anaphylaxis - were deemed unlikely given the timing, vital signs, and lack of response to fluids, vasopressors, and antihistamines. With ongoing hypotension, an intra-aortic balloon pump was inserted, achieving a systolic pressure of 100 mmHg. Laboratory tests revealed a hemoglobin drop from 13.2 g/dL pre-procedure to 11 g/dL. To rule out a retroperitoneal bleed, an urgent CT scan was performed.

Boundary: this artifact ends before the CT result. Retrieval must not assume or append a final diagnosis.